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WEB PAPER

# Preparing medical students for clinical decision making: A pilot study exploring how students make decisions and the perceived impact of a clinical decision making teaching intervention

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## Abstract

**Background:** Junior doctors are frequently faced with making difficult clinical decisions and previous studies have shown that they are unprepared for some aspects of clinical decision making.

**Aim:** To explore medical students' feelings and strategies when responsible for making clinical decisions and to obtain students' views of the effectiveness of a clinical decision making teaching intervention.

**Methods:** A teaching intervention was developed, consisting of a clinical decision making tool, a tutorial and scenarios within a simulated ward environment. A total of 23 volunteer students participated in individual interviews immediately after their simulator sessions. The qualitative data from the interviews were analysed to identify emerging themes.

**Results:** Despite extended shadowing programmes, students feel unprepared for clinical decision making as FY1s, and lack effective decision making strategies. Experiencing complex decision making scenarios through individually orientated simulation results in students being subjectively more prepared for work as FY1s.

**Conclusion:** Students continue to feel unprepared for the responsibility of clinical decision making. A teaching intervention, including simulated individual clinical scenarios, later in undergraduate training, appeared to be useful in improving medical students' decision making, specifically in relation to making a diagnosis, prioritising, asking for help and multi-tasking, but further work is required.

## Introduction

Clinical decision making, or clinical reasoning, has been described as being a foundation of effective clinical education (Dent & Harden 2009) and is a significant aspect of physician competence (Norman 2005). Examining how clinicians make decisions can be difficult (Eva 2005), but a large clinical decision making literature base does exist. According to the literature, diagnostic errors are a significant problem (Berner & Graber 2008), and despite rapid advances in medical science, the detection of incorrect diagnoses at autopsy has not changed over the past 100 years (Lundberg 1998).

Graber (2005) found that cognitive errors were the most common causes of diagnostic errors, and descriptions of how these cognitive errors can occur have been previously outlined (Croskerry 2003; Graber 2005; Croskerry 2009).

Interestingly, the vast majority of the literature on teaching clinical decision making has focused on making a correct diagnosis. While this is certainly a very important aspect, there are other aspects to clinical decision making (Lake 2005). These include clinical prioritisation and recognising limitations and asking for help. A literature search revealed no real information on how medical graduates prioritise. The previous literature on requesting senior assistance found it to be complex, with one study concluding that it was too complex to teach (Stewart 2008).

This is particularly relevant to final year medical students, as previous studies have shown that they are unprepared for prioritising (Tallentire et al. 2011), and asking for help (Illing et al. 2008). Nevertheless, GMC (2009) guidelines expect that medical educators will include these components in undergraduate curricula, and that medical graduates are competent

## Practice points

- Current FY1 shadowing programmes do not fully prepare undergraduates for decision making in a clinical environment.
- Prioritisation teaching interventions are required.
- Perplexity faced by junior doctors in making clinical decisions could possibly be minimised by providing structured cognitive frameworks.
- Simulated individual multi-tasking experiences have the potential to make students more prepared for work.
- Such interventions are welcomed by students.

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Authorized users can download,
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| AM I | HERO | NOW? |
|---|---|---|
| **Am I** trained to make this decision?<br><br>**Am I** in possession of the knowledge, skills and competencies required?<br><br>**Am I** adversely affected by the patient (e.g. alcohol, smell etc.) or circumstances (time, stress, sleep deprivation, other clinicians involved)?<br><br>**Am I** the person who should be responsible for this decision? | Have I enough information to be making this diagnosis/decision?<br><br>**H**istory<br><br>**E**xamination<br><br>**R**esults<br><br>**O**ther possible causes; other possible options? – could it be something else, could we do something else? | Is there anything happening now that is impacting on this decision?<br><br>Have I considered the impact of this diagnosis / decision on:<br><br>**Number one** patient -<br><br>Is ITU appropriate? Is NFR/ advance directive in place? Will they be upset, have they sufficient support, will I need to arrange follow-up?<br><br>**Others** -<br><br>Family is there someone else who should be involved?<br><br>Staff – is someone identified to undertake it; have they the necessary knowledge, skill and time<br><br>**Which** is the priority action – how will this decision affect the treatment given to other patients? |

**Figure 1.** 'AM I HERO NOW' tool.

in making diagnoses, prioritising and knowing when to seek help.

Various strategies have been introduced to improve clinical decision making (Croskerry 2003), based on dual process theory and cognitive biases. These include encouraging metacognition (Croskerry 2003), cognitive autopsies (Croskerry 2005; Sullivan 2009), encouraging an analytical approach through de-biasing strategies (Croskerry 2003), and encouraging both an analytical and non-analytical approach (Norman 2009; Norman & Eva 2010). However, while these approaches have shown benefits in performing a single task (Eva et al. 2007); the impact of such strategies in a clinical setting that requires multiple and interconnected decisions to be made is uncertain.

To develop students' skills in making decisions in relation to making diagnosis, prioritisation and asking for help, a tripartite teaching intervention was developed. Analytical, non-analytical and combined approaches to decision making were incorporated in the intervention.

## Component 1

Using a structured approach has previously been shown to be helpful in a number of areas relevant to clinical decision making, e.g. the airway, breathing, circulation, disability, exposure (ABCDE) approach to clinical examination (Jevon 2010), the situation-background-assessment-recommendation (SBAR) approach to communication and handover (Featherstone 2005) and the COVER ABCD – A SWIFT CHECK for anaesthetic emergencies (Runciman et al. 2005). Therefore, a decision making tool (AM I HERO NOW) was introduced to students, to be used when faced with making a decision (Figure 1). The tool was designed to be a memorable mnemonic, incorporating multiple aspects of clinical decision making, with the aim of prompting students to avoid certain recognised cognitive dispositions to respond, such as premature closure (Croskerry 2003).

## Component 2

A tutorial session was introduced providing a theoretical understanding of the process of clinical decision making. The tutorial covered cognitive theory (dual process theory and cognitive and affective biases), prioritisation and when to ask for help, and included interactive cases scenarios.

## Component 3

As a part of this intervention, students were introduced to a simulated ward environment, where they dealt with three scenarios addressing: making a diagnosis, specifically overcoming the suggestion of an incorrect diagnosis, which is known to be a legitimate instructional technique (Eva et al. 2007); prioritisation; and recognising limitations and asking for help.

This article illustrates how medical students make decisions and examines the effectiveness of this tripartite teaching intervention in its pilot run.

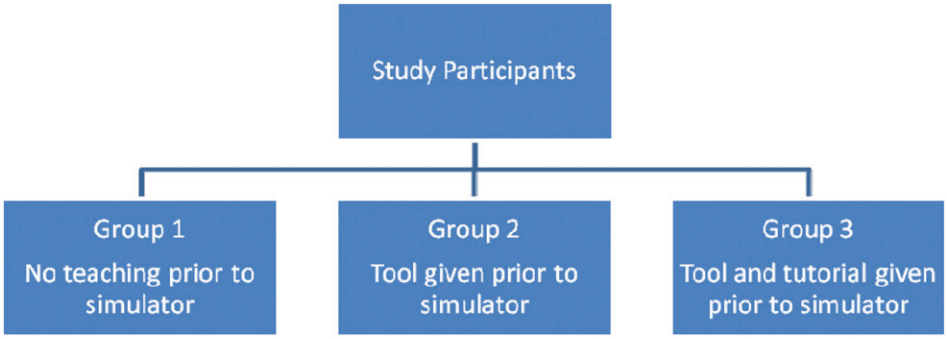


**Figure 2.** Participant Groups.

## Methodology

This study, which was considered as an educational project, was granted an ethics waiver by the NHS Research Ethics Service.

### Study approach

This was a qualitative study exploring clinical decision making and students' views of the effectiveness of a teaching intervention using data obtained from interviewing medical students.

### Study population and sampling

The sample was drawn from the population of 36 final year medical students undertaking a 6-week 'Preparation for Practice' block in the Lanarkshire region. The prospective participants had all completed their final university examinations, and this was their last undergraduate placement before they graduated and started working as junior doctors.

All 36 students were e-mailed by the researcher to explain the clinical decision making teaching programme, and to ask for volunteers to participate in interviews which would take place immediately after their simulated ward session. For those who agreed, formal written consent was obtained on the day of their simulator session, prior to the event. The consent also included permission to digitally record the interviews, and the volunteers were assured of confidentiality and anonymity of the data.

The final sample consisted of 23 volunteer students (64% of the original population of 36); 12 males and 11 females.

The students were randomly allocated into three groups – one group received no decision making teaching prior to the simulation session, one group were given the clinical decision making tool prior to the simulation session and one group were given both the tool and the tutorial prior to the simulation session (Figure 2).

### Data collection

Data were collected using real time observation of student performance and semi-structured interviews.

During the simulator, each student was followed and observed by a senior doctor. The observer completed three objective structured clinical examination style checklists (Appendix) consisting of 20 points relating to the student's performance in each of the three decision making scenarios outlined above.

The observers consisted of senior medical staff – three clinical teaching fellows and two staff grade doctors with an interest in medical education.

Within 30 min of completing their simulator session, each student was interviewed by the researcher (Calum McGregor). The interviews started off with open-ended questions about how the students experienced the simulation, which were followed by specific questions on the following areas:

- Students' thoughts on the tool and tutorial.
- Students' thoughts on existing tools (SBAR and ABCDE).
- Students' thoughts on the simulator session.
- What did students do when faced with decisions to make in the three scenarios outlined above – i.e. making a diagnosis, prioritising and deciding when to call for help? Why did they do that? How did they feel?
- What strategies did students use for decision making in the scenarios?
- What decision making training had they had prior to this event?

The researcher received each student's checklists prior to their interview, thus enabling the researcher to see what courses of action the student had taken in each scenario, and therefore direct questions accordingly during the interviews.

The interviews lasted from 14 to 50 min. All interviews were digitally recorded, transcribed and given a unique coding number.

### Data analysis

Qualitative data analysis was conducted manually using a constant comparison approach. Each transcript was read sentence by sentence, and reread by an independent researcher, to identify areas of meaning. These sections of meaning were coded and compared with codes in other transcripts.

## Results

How do medical students feel, and what strategies do they use, when making clinical decisions with regards to diagnosis, prioritisation and asking for help?

## Diagnosis

When faced with diagnosing unwell patients, most students used the ABCDE approach for the initial assessment. The students were familiar with this approach, having been taught it repeatedly during their time in medical school.

*SC1 – ABCDE is really good. I think it's good if you have no idea what is going on, it gives you something to start off with*
*SB6 – I started my systematic approach so like that kind of gave me time to think about things, by going through ABC.*
*SB5 – ABCDE gives you something useful to do as soon as you get there.*

Having a structured approach was re-assuring for students, and helped to alleviate their concerns that they would miss something important.

*SA2 – It's reassuring cause it kind of covers your back, almost like a proforma you go through and if you do everything you are not going to miss anything*

Once students had completed their ABCDE assessments, many of them struggled with what to do next, particularly if they had not made a diagnosis by this point. There was a contrast between their slick, structured ABCDE assessment and what followed after it.

*SA2 – I didn't know what to do after the ABCD. Everything kind of goes out your head*

Students were comfortable taking histories and performing ABCDE assessments. However, when faced with a confused patient who was unable to provide a history, students struggled to know how to obtain useful information, and struggled to produce a differential diagnosis.

*SA2 – I was just kind of oh no what do I do? In terms of producing a differential I am not used to doing that at all really*

## Prioritisation

Having to make decisions on how to prioritise was completely new to the students, and they had not received previous training on prioritisation. Importantly, the students recognised that the simulator exercise was the first time that they had been asked to make these types of decisions.

*SC2 – I've never really been left to make decisions and prioritise things.*
*SC2 – I am glad I had a chance to be exposed to that before I am put in the real situation. It's something I had never really done.*

Some indicated frustration that despite having completing their medical training, they had never had this opportunity before.

*SA2 – When you are a medical student you are kind of cocooned, wrapped up in cotton wool*

During their simulation session, the students were faced with multiple tasks and acutely unwell patients, and therefore had to prioritise. Students found this stressful.

*SA3 – It was stressful. When you have a sick patient and then the page goes off, that was stressful.*

When given patients to prioritise between, some students struggled to know what to ask over the phone, and lacked strategies for prioritising effectively.

*SE3 – I didn't ask for any details over the phone.*
*SA4 – I didn't get a sense of how unwell he was.*

In addition, as some students did not ask for information over the phone, they were making judgements on prioritising patients based on very limited amounts of information.

*SE3 – I don't think I asked for any info over the phone, I maybe should have asked for a bit more.*

Some strategies used for prioritising by the students were subjective or inadequate. Examples of inadequate prioritisation strategies included avoiding patients where students were unsure of the diagnosis, and attending requests to see patients in a strict chronological order rather than based on clinical need.

*SD2 – I saw them first because I kind of knew what to do with them.*
*SC1 – I saw whoever paged me first.*

The students' difficulties in prioritising were exacerbated by the fact that they had never been paged before, and were unfamiliar with pagers.

*SE1 – I thought I could ignore the pager if I was doing something else, I'm sure they will page someone else.*

## Recognising limitations and asking for help

Students were good at recognising their limitations when they encountered something that they could not do, and when they did not know what to do next, and were not afraid to ask for help in those situations.

*SC1 – I asked my senior because I felt that I had done everything in my power. I was starting to feel out of my depth.*

However, the students did not ask for help when they had two acutely unwell patients to see at the same time, or when they had multiple tasks to do. This problem was exacerbated by students being unfamiliar with the roles of other members of the on-call team.

*SA5 – I never really thought that this would be an excellent point to ask for help. I didn't know the role of a senior – I thought they were just there in case things really took a turn for the worst.*

During the simulator session, the SBAR tool was written down next to the phones. Making it readily available increased

the use of SBAR. The students felt more comfortable using the structured, SBAR approach when it came to actually speaking to their senior on the phone.

*SB2 – The SBAR structure was really useful. It took a bit of the fear away definitely*
*SC6 – I didn't use SBAR first time round, I just made a panicked phone-call, but 2nd time round SBAR was on the wall beside the phone so I just thought I will go through that and it worked.*

When students did not use the SBAR tool, some ran into problems and started feeling stressed.

*SE6 – I wish I had used SBAR. I felt a bit flustered without it. I sort of forgot.*

Did the clinical decision making teaching intervention (decision making tool, tutorial and simulator session) have any effect on students' clinical decision making?

### Decision making tool

Despite being unsure of what to do after completing their ABCDE assessment, the students did not use the 'AM I HERO NOW' decision making tool. There were three main reasons why the students did not use it; they felt it would take too long to use, it was too complex and they were unfamiliar with it.

*SC1 – You don't have time to sit there and think about every single question on that piece of paper.*
*SD3 – I thought there was a lot of information on it. Maybe a bit too much.*
*SB5 – I had never used it before. When I went in, the aide memoire went out the window.*

Despite not using it in the simulated ward session, students appreciated that using a tool such as 'AM I HERO NOW' for decision making could be useful. They liked the idea of having a structure to use when faced with a difficult decision, in the same way that they used ABCDE for assessment and SBAR for referrals to their seniors. The experience of not using a decision making tool appeared to increase its perceived value to the students.

*SD1 – Having come out actually I would have done the HERO bit especially, I think it would have been good. I think it would have been pretty useful.*

### Tutorial

The interviews revealed that the tutorial group were able to identify some of the concepts from the tutorial within the simulator session, but there was no discernable difference in the way they described their approaches to decision making.

*SD4 – The tutorial was useful. It was stuff I'd never heard of, and the ways you are influenced into decisions with stuff I had never heard of, but having it pointed out to you is probably a good thing even if you don't take much else from it. Just someone saying be aware of it is quite useful.*
*SE2 – I thought it was useful, I thought it was quite interesting. I suppose the nurse saying the patient is having a stroke would be one of the things from the tutorial.*

However, some students did not see the relevance of being taught clinical decision making theory such as dual process theory and cognitive dispositions to respond.

*SE4 – I remember the type 1 and type 2 stuff. I got it, I understood it but I don't think I found it relevant to me.*
*SE5 – The clinical based stuff you were talking about was good, but the sort of theory behind it, I don't know if it would benefit me that much.*

### Simulator session

The students found the simulator session more useful than the tutorial and the decision making tool.

*SE2 – You get more out of doing the practical stuff. I don't think there is a substitute for doing that. There is a difference between knowing what to do and actually putting it into practice.*

## Evidence of learning

It was evident from the interviews that the students had learned from the mistakes they had made during the simulator session across the domains of diagnosis, prioritisation and when to ask for help.

### Diagnosis

The simulation session helped students to understand the importance of making sure that they have enough information to arrive at a diagnosis, and to consider other possibilities.

*SE5 – The nurse said she thought the patient had had a stroke so I just took that as being gospel. I think today highlighted the need for taking a more open approach.*

### Prioritisation

During the interviews students were able to identify weaknesses in the prioritisation strategies they used in the simulator session.

*SD3 – I should probably gather more information before I decide who needs to be seen. I didn't ask for observations or anything which I should have*

Students indicated that they had learned how to prioritise more objectively.

*SA1 – I'd want to know their routine observations over the phone.*

### Recognising limitations and asking for help

After the simulator session, the students had a better understanding of when to ask for help.

*SA3 – I should have phoned someone else to come and help me instead of finishing with what I was dealing with and then going to the other one.*

The students felt more prepared for work after the simulator session.

*SE6 – Today put you in the role as the FY1 rather than just shadowing an FY1. You are actually making the decisions on the run, rather than asking someone else to make the decisions for you. It's good to do it in a simulated environment before you do it for real.*

## Other findings

Students' behaviours suggested that they are unprepared for complex decision making.

There were a number of areas where students found difficulty with decision making, especially in making decisions alone regarding diagnosis, prioritisation and asking for help, and in making simultaneous decisions.

### Simultaneous decision making (multi-tasking)

During the simulator session, many students did not write down the tasks they were asked to do. They had not considered that they might not remember them all, and therefore they forgot about a number of the tasks.

*SC6 – The phone went and she was still asking me what I thought and I was like, 'one of you give me a break for a minute while I try to think clearly.' I didn't write down what the person was saying to me over the phone, so then I forgot what they had said. Then I was getting frustrated, 'Did you actually speak to me? Oh $*@* I forgot about that!'*

Furthermore, if the students were distracted or interrupted while performing a task, they tended to lose their focus and abandon their structured approach.

*SA3 – When I was going to the patient I was paged, so I forgot to finish the ABCDE. I didn't really assess the patient properly.*

The students put these errors down to the fact that they have never been in that situation before.

*SA2 – On Preparation for Practice I have my FY1 and she makes all the decisions and I just follow her around*

Students felt that shadowing FY1s had not fully prepared them for making decisions as FY1s.

*SE6 – We've been shadowing FY1s but they are the ones making decisions and they are the ones who are stressed out.*

### Group comparison

There were no clear differences between the three groups. In addition, the number of students involved in the study was too small to make any significant comment on whether or not the tutorial helped the students to overcome the cognitive bias in the first scenario, i.e. the suggestion of an incorrect diagnosis.

## Discussion

The key findings of this study are discussed in two sections; students' decision-making and the teaching intervention.

### Students' decision making

Medical students lack effective prioritisation strategies, and described current prioritisation teaching as being extremely limited. In addition, students' behaviours in simulator scenarios suggest they are unprepared for complex decision making such as prioritisation and multi-tasking. Current FY1 shadowing programmes including group-based simulations do not appear to fully prepare undergraduates for work as an FY1, particularly with regards to their individual ownership of complex clinical decisions. However, experiencing complex, individual decision making situations through simulation has the potential to make students more prepared for work as FY1s.

Students had little, if any, experience of making a number of decisions that they will be expected to make when they start work, despite the introduction of a prolonged FY1 shadowing programme. Shadowing programmes have a number of benefits; resulting in students feeling more prepared for work (Cave et al. 2009), gaining familiarity with their work environment (Jones et al. 2006) and improving their confidence with prescribing medications (Medical Education England 2010). However, FY1 shadowing may be limited in terms of providing students with meaningful exposure to the responsibility for decision making, and in preparing them for dealing with the multiple tasks that they will be asked to do as FY1s.

The students had not previously experienced situations where they had been asked to prioritise between patients, which they will have to do when they start work. This is consistent with previous research which showed that students are not prepared for practice in the areas of prioritisation (Tallentire et al. 2011) and knowing when to request help (Illing et al. 2008).

The lack of literature and discussion around prioritisation is not unique to medicine. Lake (2005) found that, 'effective nursing prioritisation of the patient need for care is integral to daily nursing practice but there is no formal acknowledgement or study of this concept'.

The results from this study suggest that prioritisation teaching interventions need not be complex, and improvements in students' prioritisation strategies may be achievable using a simulator. Simple measures, such as teaching medical students to ask for a patient's observations over the phone, or telling students to keep a list of tasks to be done, may turn out to be effective. More work is required in this area.

The students' difficulties when distracted from a task by an interruption, or when given multiple tasks to do, are perhaps natural and understandable. However, these are situations that they will frequently encounter when they start work as FY1s, and yet the students rarely, if ever, encounter these situations during current undergraduate programmes.

Recent studies have demonstrated why doctors need to multi-task. Tipping et al. (2010) found that for 16% of hospital doctors' time, more than one event was occurring simultaneously (i.e. multi-tasking). In addition, Weigl et al. (2011) found that the frequency of hospital doctors' workflow interruptions was high. Telephones and bleepers were the most frequently recorded type of workflow interruption.

A recent paper by Eva (2009) has suggested that in the domain of clinical decision making, mistakes are necessary evils when trying to induce learning. In our study, it was evident from the interviews that students were making such mistakes in a safe, simulated environment. Individual simulation exercises such as the one in this study may enable medical students to improve their decision making without jeopardising the care of real patients.

### Teaching intervention

*The tool.* Medical students perceive a benefit from using the structured approaches ABCDE and SBAR when assessing patients and asking for help, respectively. Furthermore, the concept of a clinical decision making tool which encourages a structured approach to decision making is welcomed by students. Students found that existing tools, with which they were familiar, were very useful and helped to reduce their stress levels. It has been shown that high levels of stress can interfere with cognitive functions (Bourne & Yaroush 2003), therefore if a decision making tool could be applied, this could be very beneficial.

However, the 'AM I HERO NOW' tool, although introduced, was not actually used by students as a decision making tool during their simulated practice in this study.

With hindsight, students viewed a structure or tool for decision making as potentially useful. It is therefore possible that the 'AM I HERO NOW' tool would have been used by the participants if it had been introduced, taught and demonstrated to students at an earlier stage in their curriculum, using a more interactive format, similar to the way the ABCDE approach or SBAR is taught. A main reason for not using 'AM I HERO NOW' appeared to be lack of familiarity and so the concept of a decision making tool may yet be worthwhile.

*The tutorial.* The impact of teaching decision making cognitive theory using this study's model is not clear.

Students could remember concepts from the tutorial when interviewed, and some did spot the cognitive bias which had been included, i.e. the incorrect diagnosis in the confused patient scenario. Whether these findings constitute a measurable, beneficial impact on the students' approach to decision making is difficult to know, as some of the students in the other group were able to overcome the suggested incorrect diagnosis without having the tutorial.

A significant difficulty with delivering a tutorial is the uncertainty from the literature on how students should be taught to make clinical decisions. Some authors encourage raising awareness of cognitive dispositions to respond (CDRs) (Croskerry 2005) and adopting metacognitive, analytical approaches to problems (Elstein & Schwarz 2002), while others (Norman 2009; Norman & Eva 2010) specifically encourage teaching both analytical and non-analytical approaches. Future studies may give conclusive answers to the question of which approach works best for junior doctors, but this remains a very complex area.

*The simulation session.* The students were overwhelmingly positive about the individual simulation session, and showed evidence of quickly adapting and learning how to prioritise and make referrals. The students gained confidence from the simulation and felt that they were more prepared for working as FY1s afterwards. Specifically, they acknowledged an improved understanding, as compared with any learning to date, of their personal responsibility for decision making.

However, it remains to be seen whether these findings translate into better judgements regarding when to ask for help once the students start work, especially given the number of influences involved in making clinical decisions (Stewart 2008). The students felt that the simulator session was more beneficial than the tutorial or the decision-making tool.

## Limitations

In this study, the researcher that conducted the interviews also delivered the tutorial and the tool to the students. The participants may have therefore been biased towards giving answers that they perceived the researcher wished to hear. However, triangulation of interview data with real time observation during the simulator may have reduced this bias.

There may have been a selection bias, as only volunteers from a group of final year medical students from a single medical school took part in the study. However, we conducted this as a pilot study and future research will determine the generalisability of the findings. Although the number of participants is acceptable for a pilot study of qualitative nature, the credibility may have been increased with larger number of participants. In this study, students' perceptions of the teaching intervention were explored to assess its effectiveness. While this has yielded findings of note, an additional objective means of assessing students' performance, if available and applicable to this setting, would have given more reliable findings.

In this study, we assumed that all participating students had the same level of knowledge. However, lack of knowledge of a particular topic may have influenced individual students' clinical decision making.

## Future work

The research team are now exploring the practicalities involved in extending this study to a much larger cohort in 2012 and in obtaining longitudinal evaluation of benefit from the initial study group. Using a larger cohort would enable a

more meaningful statistical comparison of quantitative data which would minimise some of the limitations of this pilot study outlined above.

## Conclusions

In this pilot study, we have demonstrated that medical students remain not fully prepared for working as junior doctors. Awareness of, and preparedness for, the doctor's personal responsibility for decision-making were key components that were lacking. Students need further guidance on how to make key clinical decisions as junior doctors, especially in relation to making a diagnosis, prioritisation, and asking for help. Although clinical decision making is a complex process, certain 'teaching' interventions, such as using structured approaches and individual simulation sessions can be effective. However, further studies are necessary to examine the impact of these educational interventions on day-to-day clinical practice.

***Declaration of interest:*** The authors report no declarations of interest.

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# Appendix

## Checklists for simulator session

**STUDENT NAME**

**Assessment and Treatment of Chest Pain Scenario**

Record if done

**1) Introduces self to patient. Check/Assess the airway (A)**
Notes Patient able to talk therefore airway patent.
Assesses for tracheal deviation
Asks patient to report if in pain. (Central Chest pain radiating to L arm) Explain actions and treatment.
**2) Check/Assess breathing (B)**
Checks Respiratory rate
Checks if Breathing symmetrical
Assesses air entry – Bilateral air entry. No abnormal sounds
Checks for Cyanosis
Commences High Flow 02 via non-rebreathing mask as no history of Type II respiratory failure – info given to students
Requests/attaches SPO2
**3) Check/Assess circulation (C)**
Looks for Diaphoretic?
Checks Capillary Refill Time
Checks Pulse Rate
Monitoring Attach/request cardiac monitoring heart rate
Checks BP, attach request blood pressure cuff
Listens to heart sounds
Requests/Explains need for 12 lead ECG
**4) Diagnosis, Treatment and Management**
Inserts IV access
Correctly interprets information from history, exam and ECG findings – i.e. diagnoses acute coronary syndrome *
Initiates treatment for ACS MONA (Morphine, Oxygen, Nitrate, Aspirin) *
Recognises limitations and asks for help *

Marker's Signature _____
Marker's Name (print)_____

**STUDENT NAME**
**Confused Patient Scenario**

Record if done

**1) Introduces self to patient. Check/Assess the airway (A)**
Notes Patient able to communicate, therefore airway patent.
**2) Check/Assess breathing (B)**
Checks respiratory rate
Checks if breathing symmetrical?
Assesses air entry
Requests/Attaches Sats
Commences O2 where necessary
**3) Check/Assess circulation (C)**
Checks Capillary Refill Time
Checks Pulse Rate
Checks BP
**4) Disability D**
Notes GCS
Undertakes Neuro Exam
**5) Diagnostic Sieve before diagnosis**
Considers alternatives - Looks for rash *
Considers alternatives – Looks for evidence of head injury *
Looks at drug chart *
Attempts to obtain more info (notes/nursing staff) to ascertain patient's normal level of functioning *
Asks for BM without prompting *
**6) Diagnosis and Management**
Treats BM appropriately *
Thinks about measures to prevent further hypo after correction (regular BMs, reduce insulin, long-acting carbohydrate etc.) *

Marker's Signature _____
Marker's Name (print)_____

* represented the actions most relevant to clinical decision making

**STUDENT NAME**
**Prioritisation Scenario Hypotensive Man**

Record if done

**1) Introduces self to patient. Check/Assess the airway (A)**
Notes Patient able to talk so airway patent.
Assesses for tracheal deviation as potentially fatal sign
**2) Check/Assess breathing (B)**
Checks Respiratory rate
Checks Breathing symmetrical?
Assesses air entry
Commences High Flow 02 via non-rebreather mask as no history of Type II respiratory failure
Requests/attaches SPO2
**3) Check/Assess circulation (C)**
Checks Capillary Refill Time
Checks Pulse Rate
Checks BP
Listens to heart sounds
Checks for peripheral oedema
Looks for signs of haemorrhage around bed *
**4) Disability D**
Comments on GCS or AVPU
**5) Management**
Correctly identifies hypotensive patient as high priority *
Correctly prioritises urgent nature of situation and attends patient *
Asks for/looks at fluid balance chart *
Asks for/looks at drug chart *
Send bloods for FBC, U and Es etc
Asks for ECG *
Inserts IV access if not already obtained
Gives fluid challenge quickly *
Re-assesses patient post fluid challenge*

Marker's Signature _____
Marker's Name (print)_____

* represented the actions most relevant to clinical decision making

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